Partial Results on Sun's A179524 Divisibility Conjecture

A quadratic invariant, prime-square transfer, and complex multiplication

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Abstract

Let $a_n = \sum_k (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$ and $S(n) = \sum_{j=0}^{n-1} (20j+17)a_j$. Sun conjectured in 2010 that $n \mid S(n)$ for every positive integer n. We do not prove that full weighted assertion. We do prove the complete inert unweighted branch, namely that the sum of the first p terms is zero modulo p^2 for $p \equiv 11, 13, 17, 19 \pmod{20}$, using a quadratic differential invariant. We also prove $\mathrm{rad}(n) \mid S(n)$ and the exact identity $v_2(S(n)) = v_2(n)$ for every n. A first-order digit expansion gives $S(p^2 m) \equiv U(m) S(p^2) \pmod{p^2}$, where $U(m) = \sum_{j<m} a_j$, for every odd prime p and every positive m. Using a CM elliptic curve of discriminant -20, we prove $p^2 \mid S(p^2 m)$ whenever $p \equiv 11, 13, 17, 19 \pmod{20}$. We also give a symbolic proof for $p = 3$. Exact computation supplies the same all-cofactor conclusion for every odd prime $p \le 67$. Consequently the full conjecture holds on an explicit set of integers of natural density at least 0.9481679114. This density measures a proved domain, not the proportion of the mathematical difficulty resolved. The remaining higher odd prime powers, experimental strengthening, and limits of the priority investigation are stated explicitly.

1 The unchanged question and the proved domain

Use the convention that a binomial coefficient with an inadmissible lower index is zero. Define

$$M(n,k) = \binom{n}{k}\binom{n-k}{k}, \qquad a_n = \sum_{0 \le 2k \le n} (-4)^k M(n,k)^2, \\ S(n) = \sum_{j<n} (20j+17)a_j. \tag{1}$$

For $0 \le 2k \le n$, the factorial expression is $M(n,k) = n!/[k!^2(n-2k)!]$. The initial values are $1, 1, -15, -143, 1, 12801, 100401, -555855$. The unchanged assertion is

$$n \mid S(n) \qquad (n \ge 1). \tag{2}$$

This is Conjecture 5.8(ii), printed p. 33, of Sun's 2010 preprint [1], with $a_n = W_n(1,-4)$. The OEIS record A179524 attributes it to 1 July 2010 [2]. The latest version of arXiv:1006.2776 was withdrawn to merge material into other papers; the numbered source cited here is explicitly version 11.

For an integer $N \ne 0$, $v_p(N)$ denotes its p-adic valuation. Let $\mathcal{P}_0$ be the eighteen odd primes at most 67, and put

$$\mathcal{P}_* = \mathcal{P}_0 \cup \{p \text{ prime} : p \equiv 11, 13, 17, 19 \pmod{20}\}.$$

Define $e_p = 2$ for $p \in \mathcal{P}_*$ and $e_p = 1$ for other odd primes.

Theorem 1.1 (Combined partial result)**.** *For every positive integer n,*

$$v_2(S(n)) = v_2(n), \qquad 2^{v_2(n)} \prod_{\substack{p\mid n\\ p \text{ odd}}} p^{\min(v_p(n),e_p)} \mid S(n). \tag{3}$$

In particular, (2) holds if $v_p(n) \le e_p$ for every odd prime p. There is no restriction on the exponent of 2.

The prime cases in this theorem use Sun's established Ramanujan-type supercongruence [3]. The inert-prime-square cases use standard CM reduction and the Hasse invariant. The finite set $\mathcal{P}_0$ uses the exact certificate in Appendix A, together with an all-cofactor transfer theorem. No computational bound is imposed on the cofactor m in any of these conclusions.

Corollary 1.2 (An explicit density lower bound)**.** *The conjecture holds on a set of natural density at least*

$$\frac{8}{\pi^2} \prod_{\substack{3\le p\le 67\\ p \text{ prime}}} \frac{1-p^{-3}}{1-p^{-2}} = \frac{8}{\pi^2} \prod_{\substack{3\le p\le 67\\ p \text{ prime}}} \frac{p^2+p+1}{p(p+1)} = 0.94816791146370398749\ldots.$$

Proof. Take the smaller set defined by $v_p(n) \le 2$ for odd $p \le 67$ and $v_p(n) \le 1$ for $p > 67$, with arbitrary $v_2(n)$. The Euler product for these valuation restrictions is the displayed expression. Its existence follows by first imposing finitely many local conditions, using the Chinese remainder theorem, and bounding the omitted tail by $\sum_{p>y} p^{-2}$. The additional inert-prime cases can only enlarge the proved set. □

2 Prime divisibility and Lucas propagation

We first record two integer identities:

$$\binom{j}{m}^2 = \sum_{i=0}^{m} \binom{m}{i}\binom{m+i}{i}\binom{j}{m+i}, \tag{4}$$

$$\sum_{j<n} (20j+17)\binom{j}{t} = (20t+17)\binom{n}{t+1} + 20(t+1)\binom{n}{t+2}. \tag{5}$$

The first counts ordered pairs of m-subsets by their union size. The second follows from the hockey-stick identity and $j\binom{j}{t} = t\binom{j}{t} + (t+1)\binom{j}{t+1}$.

Lemma 2.1 (Ramanujan reduction)**.** *For an odd prime p, $h = (p-1)/2$, and $d = \lfloor p/4 \rfloor$,*

$$S(p) \equiv -(-1)^h \sum_{j=0}^{d} (20j+3)\frac{(4j)!}{j!^4(-1024)^j} \pmod p. \tag{6}$$

Proof. For $0 \le k \le h$, the polynomial $F_k(X) = \binom{X}{2k}^2$ is defined over $\mathbb{F}_p$ and satisfies $F_k(2k-1-X) = F_k(X)$. Thus

$$\sum_{r\in\mathbb{F}_p} (20r+17)F_k(r) = (20k+7)\sum_{r\in\mathbb{F}_p} F_k(r).$$

Apply (4) and sum over $0 \le r < p$. Modulo p, the only surviving binomial $\binom{p}{2k+i+1}$ has $2k+i = p-1$. Since $i = p-1-2k$ is even, its coefficient reduces to $\binom{2k}{p-1-2k}$. Therefore

$$S(p) \equiv \sum_{k=\lceil h/2\rceil}^{h} (20k+7)(-4)^k \binom{2k}{k}^2 \binom{2k}{2h-2k} \pmod p.$$

Put $k = h - j$. The needed congruences are

$$20k + 7 \equiv -(20j + 3), \quad (-4)^{h-j} \equiv (-1)^h(-4)^{-j},$$

$$\binom{p-1-2j}{h-j}^2 \equiv \binom{h+j}{2j}^2 \equiv \frac{\binom{2j}{j}^2}{256^j}, \qquad \binom{p-1-2j}{2j} \equiv \binom{4j}{2j}.$$

For the middle identity use $\binom{j-1/2}{2j} = (-1)^j\binom{2j}{j}/16^j$. All denominators are units modulo p. These formulas give (6). □

Sun's Theorem 1.2 [3] states, for $p > 3$,

$$\sum_{j=0}^{h}(20j+3)\frac{(4j)!}{j!^4(-1024)^j} \equiv (-1)^h p(2^{p-1} + 2 - (2^{p-1}-1)^2) \pmod{p^4}.$$

For $d < j \le h$, the numerator $(4j)!$ is divisible by p and $j!$ is not. Hence Lemma 2.1 gives $p \mid S(p)$. The remaining primes satisfy $S(2) = 54$ and $S(3) = -801$.

Lemma 2.2 (Digit congruence)**.** *For every odd prime p, $u \ge 0$, and $0 \le r < p$,*

$$a_{pu+r} \equiv a_u a_r \pmod{p}.$$

Proof. Write the inner index as $k = pv+s$, $0 \le s < p$. Multinomial Lucas makes $M(pu+r, pv+s)$ vanish modulo p unless $2s \le r$. In the latter case

$$M(pu + r, pv + s) \equiv M(u, v)M(r, s) \pmod{p}.$$

The high-digit range is exactly $0 \le 2v \le u$. Also $(-4)^{pv+s} \equiv (-4)^{v+s} \pmod{p}$. Summing the factored terms proves the claim. □

For every $m \ge 1$, it follows that

$$S(pm) \equiv S(p) \sum_{u<m} a_u \equiv 0 \pmod{p}. \tag{7}$$

The case $p = 2$ follows directly from $a_u \equiv 1 \pmod 2$. Consequently $\mathrm{rad}(n) \mid S(n)$ for every n. This argument alone does not increase the prime-power exponent.

3 The entire 2-part, with exact valuation

Theorem 3.1. *For every $n \ge 1$, $S(n)/n$ is an odd 2-adic integer. Thus $S(n) \ne 0$ and $v_2(S(n)) = v_2(n)$.*

Proof. Rewrite $M(j,k) = \binom{j}{2k}\binom{2k}{k}$. For a fixed $k \ge 1$, use (4) with $m = 2k$, followed by (5), and divide by n. Its contribution is an integer linear combination of

$$(-4)^k\binom{2k}{k}^2\left(\frac{20t+17}{t+1}\binom{n-1}{t} + \frac{20(t+1)}{t+2}\binom{n-1}{t+1}\right), \qquad 2k \le t \le 4k.$$

The prefactor has 2-adic valuation at least $2k+2$, because $\binom{2k}{k} = 2\binom{2k-1}{k-1}$. Both displayed denominators are at most $4k+2 < 2^{2k+2}$. Every such term therefore has valuation at least one. All sums are finite. The $k = 0$ contribution is

$$\frac{1}{n}\sum_{j<n}(20j+17) = 10n + 7,$$

which is odd. This proves the assertion, including $n = 1$. □

4 A prime-square transfer valid for every cofactor

Put $f(n,k) = (-4)^k M(n,k)^2$, and define

$$L_n = \sum_k k f(n,k), \qquad U(N) = \sum_{n<N} a_n, \quad T(N) = \sum_{n<N} n a_n, \quad V(N) = \sum_{n<N} L_n.$$

Thus $S(N) = 20T(N) + 17U(N)$. For an odd prime p, let

$$Q_p = \frac{(-4)^{p-1} - 1}{p},$$
$$A_r = \sum_s f(r,s)(H_r - H_{r-2s}), \qquad B_r = \sum_s f(r,s)(H_{r-2s} - H_s) \quad (0 \le r < p),$$

where $H_0 = 0$, $H_j = \sum_{i=1}^{j} 1/i$, and only admissible s are included. These harmonic numbers are p-integral.

Lemma 4.1 (First-order digit expansion)**.** *For every $q \ge 0$ and $0 \le r < p$,*

$$a_{pq+r} \equiv a_q a_r + p\{2q a_q A_r + L_q(Q_p a_r + 4B_r)\} \pmod{p^2}. \tag{8}$$

Proof. Let $E_p(N) = \prod_{1 \le i \le N,\ p \nmid i} i$. The exact factorization $N! = p^{\lfloor N/p \rfloor} \lfloor N/p \rfloor! E_p(N)$, together with $\sum_{i=1}^{p-1} i^{-1} = 0$ in $\mathbb{F}_p$, gives

$$E_p(pq+r) \equiv ((p-1)!)^q r!(1 + pqH_r) \pmod{p^2}.$$

Write $k = pj + s$. If $2s > r$, Lucas gives $p \mid M(pq+r,k)$, so its square vanishes modulo p^2. Otherwise the factorial factorization gives

$$M(pq+r, pj+s) \equiv M(q,j)M(r,s)[1 + p\{qH_r - 2jH_s - (q-2j)H_{r-2s}\}] \pmod{p^2}.$$

Only E_p-factors and small factorials are inverted. The integer $M(q,j)$ is retained exactly, so higher carries cause no problem. Finally

$$(-4)^{pj+s} \equiv (-4)^{j+s}(1 + pjQ_p) \pmod{p^2}.$$

Square the multinomial congruence, multiply, and sum over j, s. □

Theorem 4.2 (Prime-square transfer)**.** *For every odd prime p and every positive integer m,*

$$S(p^2 m) \equiv U(m) S(p^2) \pmod{p^2}. \tag{9}$$

Proof. Write $w_r = 20r + 17$, $U_p = U(p)$, $T_p = T(p)$, and $V_p = V(p)$. Summing (8) with weight $20(pq+r) + 17$ gives

$$S(pM) \equiv S(p)U(M) + pC_p T(M) + pD_p V(M) \pmod{p^2}, \tag{10}$$

where

$$C_p = 20U_p + 2\sum_{r<p} w_r A_r, \qquad D_p = 4\sum_{r<p} w_r B_r.$$

The omitted term is exactly $pQ_p S(p)V(M)$; it vanishes because $p \mid S(p)$, proved in Section 2. Lucas also gives $L_{pu+r} \equiv a_u L_r \pmod{p}$, since the index $pv + s$ reduces to s. Hence

$$U(pm) \equiv U_p U(m), \quad T(pm) \equiv T_p U(m), \quad V(pm) \equiv V_p U(m) \pmod{p}.$$

Substitute $M = pm$ into (10). Its three coefficients $S(p), pC_p, pD_p$ are divisible by p, so these substitutions are valid modulo p^2. The result is

$$S(p^2 m) \equiv U(m)[S(p)U_p + pC_p T_p + pD_p V_p].$$

The bracket is $S(p^2)$ modulo p^2 by (10) with $M = p$. □

Remark 4.3 (What a finite base verifies). For a fixed prime p, an exact verification of $p^2 \mid S(p^2)$ now proves $p^2 \mid S(p^2 m)$ for *every* m. It is a proof for an unbounded cofactor, but not a verification over all primes. The theorem also gives no p^3 conclusion.

5 An infinite prime-square family from complex multiplication

Theorem 5.1. *If $p \equiv 11, 13, 17, 19 \pmod{20}$ is prime, then*

$$p^2 \mid S(p^2 m) \qquad (m \geq 1).$$

Proof. For an indeterminate c, define

$$W_r(c) = \sum_k c^k M(r,k)^2, \qquad \mathcal{U}_p(c) = \sum_{r<p} W_r(c).$$

Set $h = (p-1)/2$ and $d = \lfloor p/4 \rfloor$. The binomial-square summation in Section 2, followed by $k = h - j$, gives the polynomial identity

$$\mathcal{U}_p(c) \equiv c^h \sum_{j=0}^{d} \frac{(4j)!}{j!^4 (256c)^j} \pmod{p}. \tag{11}$$

All apparent negative powers cancel. The finite Legendre identity

$$P_d(t)^2 = \sum_{j=0}^{d} \binom{d}{j}\binom{d+j}{j}\binom{2j}{j}\left(\frac{t^2-1}{4}\right)^j$$

and $\{-d, d+1\} = \{1/4, 3/4\}$ in $\mathbb{F}_p$ imply

$$\mathcal{U}_p(c) \equiv c^h P_d(t)^2, \qquad t^2 = 1 - \frac{1}{c}. \tag{12}$$

This is an identity in the parameter, not merely an equality at a particular numerical value.

Take $t_0 = \sqrt{5}/2$, corresponding to $c = -4$, and consider

$$E_{t_0}: \quad y^2 = x\left(x^2 + 2x + \frac{1-t_0}{2}\right).$$

Its invariants are

$$\Delta = 8(t_0-1)^2(t_0+1), \qquad j(E_{t_0}) = \frac{64(3t_0+5)^3}{(t_0-1)^2(t_0+1)} = 632000 + 282880\sqrt{5}.$$

Duke [4] identifies this value as $j(i\sqrt{5})$; hence the curve has complex multiplication of discriminant -20. The other sign of t_0 gives the conjugate CM value. The coefficients are integral away from 2, and $(t_0-1)(t_0+1) = 1/4$, so the model has good reduction at every prime above an odd rational prime.

The stated residue classes are exactly $\left(\frac{-5}{p}\right) = -1$. Deuring's reduction theorem therefore makes this reduction supersingular; see [5]. This holds over $\overline{\mathbb{F}}_p$, including when $t_0 \notin \mathbb{F}_p$.

For completeness, the Hasse coefficient can be read off directly. Let $b = (1-t)/2$. The coefficient of x^{p-1} in $[x(x^2+2x+b)]^h$ is

$$\sum_{k=0}^{d} \frac{h!}{k!^2(h-2k)!}\, 2^{h-2k} b^k = 2^h \sum_{k=0}^{d} \frac{(-h/2)_k((1-h)/2)_k}{k!^2} b^k \equiv 2^h P_d(t) \pmod{p}.$$

Here $(z)_k$ is the rising factorial. The standard Hasse-invariant criterion [6, Theorem V.4.1(a)] therefore gives $P_d(t_0) = 0$. Equation (12) yields

$$U_p = \mathcal{U}_p(-4) \equiv 0 \pmod{p}. \tag{13}$$

There are two elementary ways to obtain the other moments. The reflection used in Section 2 also gives, for each k,

$$\sum_{r<p} r\binom{r}{2k}^2 \equiv \left(k-\frac{1}{2}\right)\sum_{r<p}\binom{r}{2k}^2 \pmod p.$$

Thus $T_p \equiv V_p - U_p/2 \pmod p$. Since $20T_p + 17U_p = S(p) \equiv 0 \pmod p$ and $p \neq 2, 5$, (13) implies $T_p = V_p = 0$ in $\mathbb{F}_p$. Alternatively, differentiate (12): the square factor gives $\mathcal{U}_p'(-4) = 0$ because $t_0 \neq 0$; then $V_p = -4\mathcal{U}_p'(-4) = 0$.

Finally the bracket in the proof of Theorem 4.2 vanishes modulo p^2: $p \mid S(p)$ and U_p, T_p, V_p all vanish modulo p. This proves the theorem for every cofactor m. □

6 A complete unweighted branch: a quadratic invariant

The following statement is separate from the weighted all-n target. It supplies a full proof of the zero branch of Sun's Conjecture 5.8(i) [1], recorded as `OeisA179524.conjecture3` in [7]. Its historical priority is not asserted.

Theorem 6.1 (Automatic lift away from singular parameters)**.** *Let p be an odd prime and set $Y_p(c) = \sum_{r=0}^{p-1} W_r(c)$. If $c_0 \in \mathbb{Z}_p$, $c_0(c_0-1) \in \mathbb{Z}_p^\times$, and $Y_p(c_0) \equiv 0 \pmod p$, then*

$$Y_p(c_0) \equiv 0 \pmod{p^2}.$$

The binomial reductions used below are established ingredients: Sun [9, Theorem 1.1] relates this sum to a sum of Apéry polynomials, and Guo–Zeng [10, Theorem 1.3] give the binomial reduction used here. We include its direct proof to make the subsequent differential argument checkable without additional files. The finite-invariant lifting strategy itself has a direct antecedent in Babanskyy [11, Lemmas 2.1–2.2]: a quadratic differential invariant, explicit treatment of exceptional degree-p coefficients, and a double-root argument yield a lift from modulo p to modulo p^2. Here that strategy is adapted to the dual polynomial below. The proof is given directly; the invariant principle is not claimed as a new invention.

Lemma 6.2. *For an even integer m with $0 \le m \le p-1$,*

$$\sum_{r=m}^{p-1}\binom{r}{m}^2 \equiv \binom{p+m}{2m+1} \pmod{p^2}.$$

Proof. For $m \le r < p$, all denominators in $H_r - H_{r-m}$ are p-adic units, and

$$\binom{p+m-1-r}{m} \equiv \binom{r}{m}[1 - p(H_r - H_{r-m})] \pmod{p^2}.$$

The sign is positive because m is even. The involution $r \mapsto p+m-1-r$ permutes $m, \dots, p-1$, preserves $\binom{r}{m}^2$ modulo p, and reverses $H_r - H_{r-m}$ modulo p. The last fact follows from $H_{p-1-u} \equiv H_u \pmod p$. Thus the sum of the first-order corrections is zero modulo p. Vandermonde's identity gives

$$\sum_{r=m}^{p-1}\binom{r}{m}\binom{p+m-1-r}{m} = \binom{p+m}{2m+1},$$

proving the claim, including $m = 0$ and $m = p-1$. □

Lemma 6.3 (Dual polynomial and differential equation)**.** *Put $h = (p-1)/2$ and*

$$Z_p(c) = \sum_{s=0}^{h} v_s c^s, \qquad v_s = p\frac{(2s)!^4}{s!^4(4s+1)!}.$$

Then $v_s \in \mathbb{Z}_p$, $Y_p \equiv Z_p \pmod{p^2}$ coefficientwise, and, with $\theta = c\,d/dc$,

$$\mathcal{L}Y_p \equiv 0 \pmod{p^2}, \qquad \mathcal{L} = \theta(4\theta-1)(4\theta+1) - 2c(2\theta+1)^3. \tag{14}$$

Proof. The denominator $(4s+1)!$ contains at most one factor p, since $4s+1 \le 2p-1$; it is canceled by the displayed factor p. Lemma 6.2 makes the coefficient of c^s in Y_p congruent to

$$\binom{2s}{s}^2 \binom{p+2s}{4s+1} = p\binom{2s}{s}^2 \frac{\prod_{j=1}^{2s}(p^2-j^2)}{(4s+1)!}.$$

The product is $(2s)!^2$ modulo p^2, since $2s$ is even. Before division the error has valuation at least three; the factorial denominator loses at most one. Hence the error is still zero modulo p^2, proving the coefficientwise congruence.

The exact coefficient relation is

$$s(4s-1)(4s+1)v_s = 2(2s-1)^3 v_{s-1} \qquad (1 \le s \le h).$$

Consequently the entire truncation residual is

$$\mathcal{L}Z_p = -2p^3 v_h c^{h+1}.$$

It vanishes modulo p^3. Applying the integer differential operator to $Y_p - Z_p$ proves (14). No coefficient s or $4s \pm 1$ has been inverted modulo p. □

Lemma 6.4 (Quadratic differential invariant)**.** *For a polynomial Y, define*

$$\mathcal{J}[Y] = 16c^2(c-1)(Y'^2 - 2YY'') - 16c(3c-2)YY' - (4c+1)Y^2.$$

Then

$$\frac{d}{dc}\mathcal{J}[Y] = 2Y\,\frac{\mathcal{L}Y}{c}.$$

In particular, $\mathcal{J}[Y_p] \equiv 0 \pmod{p^2}$ coefficientwise.

Proof. The differential identity follows by expanding

$$\frac{\mathcal{L}Y}{c} = 16c^2(1-c)Y''' + (48c - 72c^2)Y'' + (15 - 52c)Y' - 2Y.$$

The right side is polynomial. Thus $\mathcal{J}[Y_p]'$ is zero modulo p^2. A zero derivative alone would not suffice, because a degree-p term could survive. We check that term explicitly.

Since $\deg Y_p = h$, the degree of $\mathcal{J}[Y_p]$ is at most $2h+1 = p$. Its constant coefficient is $-Y_p(0)^2 = -p^2$. If K is the leading coefficient of Y_p, its degree-p coefficient is

$$[16(h^2 - 2h(h-1)) - 48h - 4]K^2 = -4(2h+1)^2K^2 = -4p^2K^2.$$

Both endpoints vanish modulo p^2. For every intermediate index $1 \le j \le p-1$, the derivative gives $j\,[c^j]\mathcal{J}[Y_p] = 0$ modulo p^2, and j is a unit. All coefficients therefore vanish. □

Proof of Theorem 6.1. Equation (12) gives

$$Y_p(c) \equiv c^h P_d(t)^2 \pmod{p}, \qquad d = \lfloor p/4 \rfloor, \quad t^2 = 1 - 1/c.$$

Choose t_0 over the algebraic closure of the residue field with $t_0^2 = 1 - 1/c_0$. The assumptions make $t_0 \neq 0$, $t_0^2 \neq 1$, and $P_d(t_0) = 0$. Such a Legendre root is simple. Indeed, if its multiplicity were $m \ge 2$, the lowest-degree term of

$$(1-t^2)P_d'' - 2tP_d' + d(d+1)P_d = 0$$

would have a nonzero coefficient proportional to $(1-t_0^2)m(m-1)$, since $m \le d < p$. The degree-zero case has no roots and is vacuous.

Because $dt/dc = 1/(2tc^2)$, differentiating the polynomial congruence gives

$$Y_p'(c_0) \equiv 0, \qquad Y_p''(c_0) \equiv 2c_0^h P_d'(t_0)^2 (2t_0c_0^2)^{-2} \neq 0 \pmod{p}.$$

The expression is in the base residue field, as is the left side; the field extension is used only to describe the root.

Evaluate the invariant at c_0. Both $Y_p(c_0)$ and $Y_p'(c_0)$ are divisible by p, so every term except the Y_pY_p'' term vanishes modulo p^2. We obtain

$$32c_0^2(c_0-1)Y_p(c_0)Y_p''(c_0) \equiv 0 \pmod{p^2}.$$

Its coefficient multiplying $Y_p(c_0)$ is a p-adic unit. This forces $Y_p(c_0) \equiv 0 \pmod{p^2}$. □

Corollary 6.5 (Sun's inert unweighted branch)**.** *For every prime* $p \equiv 11, 13, 17, 19 \pmod{20}$*,*

$$\sum_{r=0}^{p-1} a_r \equiv 0 \pmod{p^2}.$$

Proof. The CM argument in Theorem 5.1 gives $Y_p(-4) = 0 \pmod{p}$. Here $c_0 = -4$ satisfies $c_0(c_0-1) = 20$, a unit at all the specified primes. Apply Theorem 6.1. □

The corollary closes all quantifiers of this particular unweighted branch. It neither proves the split-prime evaluation branches of Sun's Conjecture 5.8(i), nor the weighted all-n statement (2) in Conjecture 5.8(ii).

7 A direct symbolic proof modulo 9

Proposition 7.1. *For every* $m \ge 1$*,* $9 \mid S(9m)$*.*

Proof. The digit expansion (8) at $p = 3$, or a direct factorial expansion, gives for every $q \ge 0$

$$a_{3q} \equiv a_q + 6L_q, \qquad a_{3q+1} \equiv a_q, \qquad a_{3q+2} \equiv -15a_q + 3L_q \pmod{9}. \tag{15}$$

Here is an explicit check of the coefficients. The only low-digit pairs with $2s \le r$ and their corresponding weights are

$$(r,s) = (0,0), (1,0), (2,0), (2,1), \qquad f(r,s) = 1, 1, 1, -16.$$

Use $H_0 = 0, H_1 = 1, H_2 = 3/2$, together with

$$(-4)^{3j+s} \equiv (-4)^{j+s}(1+6j) \pmod{9}.$$

The correction divided by 3 is

$$2q(H_r - H_{r-2s}) + 4j(H_{r-2s} - H_s) + 2j.$$

Its q-coefficient is zero modulo 3 in all four cases, and its j-coefficients are $2, 0, 2, 1$, respectively. All low-carry terms vanish modulo 9; higher carries remain in $M(q, j)$. This proves (15) for arbitrary q.

Consequently

$$\sum_{r=0}^{2} (20(3q+r)+17)a_{3q+r} \equiv 3(qa_q + L_q) \pmod{9}.$$

Modulo 3, Lucas gives $a_{3u+r} \equiv a_u a_r$ and $L_{3u+r} \equiv a_u L_r$. Since

$$(a_0, a_1, a_2) \equiv (1, 1, 0), \qquad (L_0, L_1, L_2) \equiv (0, 0, 2) \pmod 3,$$

we have

$$\sum_{r=0}^{2}((3u+r)a_{3u+r} + L_{3u+r}) \equiv a_u \sum_{r=0}^{2}(ra_r + L_r) = 0 \pmod 3.$$

Sum the preceding modulo-9 block over $q < 3m$, and group the right side into these three-term blocks. □

8 Exact computation and an equivalent unsolved condition

Appendix A evaluates the original sums using integer arithmetic. For fixed n, its summands satisfy

$$f(n, 0) = 1, \qquad f(n, k+1) = \frac{-4(n-2k)^2(n-2k-1)^2}{(k+1)^4} f(n, k). \tag{16}$$

Every division is checked to be exact. The run verified $n \mid S(n)$ for $1 \le n \le 5000$, with no counterexample. In particular $p^2 \mid S(p^2)$ was verified for

$$\mathcal{P}_0 = \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67\}.$$

Theorem 4.2 converts these eighteen exact base checks into $p^2 \mid S(p^2 m)$ for all m, proving the finite-prime part of Theorem 1.1. This completes that theorem and its density corollary.

Define the Newton coefficients

$$b_\ell = \sum_{j=0}^{\ell}(-1)^{\ell-j}\binom{\ell}{j} a_j, \qquad b_{-1} = 0.$$

Binomial inversion and (5) give the exact formula

$$\frac{S(n)}{n} = \sum_{\ell<n}\binom{n-1}{\ell}\frac{(20\ell+17)b_\ell + 20\ell b_{\ell-1}}{\ell+1}. \tag{17}$$

Since the binomial transform and its inverse have integer coefficients, the full conjecture is equivalent to

$$\ell + 1 \mid 3b_\ell + 20b_{\ell-1} \quad (\ell \ge 0). \tag{18}$$

This is a reformulation, not a proof of the missing cases. A useful constant-term model is

$$P = (1 + x + y)\left(1 + \frac{z}{x} - \frac{4}{yz}\right), \quad Q = P - 1, \qquad a_n = \mathrm{CT}\, P^n, \quad b_n = \mathrm{CT}\, Q^n.$$

Matching the three exponents proves these identities directly.

The same exact run found no failures of the following *unproved* stronger congruences:

$$S(p^2 m) \overset{?}{\equiv} p^2 S(m) \pmod{p^{v_p(m)+3}} \quad (p \text{ odd}), \tag{19}$$

$$S(pm) \overset{?}{\equiv} pS(m) \pmod{p^{v_p(m)+2}} \quad (p \equiv 1 \pmod 4). \tag{20}$$

There were 989 tested pairs in (19), with $p^2 m \le 5000$, and 3787 in (20), with $pm \le 5000$. If (19) were proved for all its parameters, induction in steps of two, starting from (7), would settle every odd prime power and hence the entire conjecture. We have not proved it. The valid transfer (9), whose modulus is only p^2, must not be substituted for (19).

9 Status, provenance, and limits

The exact unchanged target is currently recorded as `OeisA179524.conjecture4`, with `category research open` and an unfilled proof, in the inspected formal-conjectures source [7]. The inspected path history first adds the file on 20 August 2026, not 6 November. The historical mathematical source is already the 2010 conjecture, irrespective of later lists.

Although the OEIS Open paper [8] does not mention the sequence by its identifier, its released benchmark does include it. The isolated target `oeis_a179524_sun_conjecture_2` conjoins (2) with the separate prime-square refinement

$$S(p) \stackrel{?}{\equiv} p\left(7 + 10\left(\frac{-1}{p}\right)\right) \pmod{p^2}.$$

The three inspected full-benchmark submissions did not supply accepted proofs of that conjunction. Omission from the paper's prose is therefore not evidence of omission from the benchmark.

The full repeated-odd-prime-power problem remains unresolved in this note. In particular, no general p^3 divisibility is asserted, even for the inert primes in Theorem 5.1. The additional small-prime square checks do not cover arbitrarily large split-prime squares. The separate inert unweighted branch is proved in Section 6; it must not be identified with the original weighted target.

Independent derivations checked the digit expansions, high-carry cases, the CM identification, all-cofactor arguments, and the quadratic invariant including its exceptional degree-p coefficient. This is mathematical review within the research workflow, not external peer review or Lean formalization. The finite calculations are reproducible in Appendix A. No claim is made that the partial results, their proof mechanisms, or their combination are historically first. The known ingredients are attributed above. The original full-success requirements are not all met: the full all-n quantifier is not closed, and nonduplication has not been established by a comprehensive priority review.

A Complete exact verification program

The following standard-library-only Python 3.11-or-later program is contained in this single `.tex` source. It computes the original a_n and $S(n)$, not a guessed holonomic recurrence. Its exact base calculations support the finite-prime corollary; the experimental higher lifts remain experimental. The recorded output was: no divisibility failures through 5000, no failures in 989 double-step tests, and no failures in 3787 split-prime tests.

```
from math import isqrt
from pathlib import Path
import time, json, hashlib, sys
sys.set_int_max_str_digits(0)
N=5000
start=time.monotonic()
a=[]; S=[0]
for n in range(N):
    term=1; an=1
    for k in range(n//2):
        numerator=-4*term*(n-2*k)**2*(n-2*k-1)**2
        denominator=(k+1)**4
        term,rem=divmod(numerator,denominator)
        assert rem==0
        an+=term
    a.append(an)
    S.append(S[-1]+(20*n+17)*an)
    assert S[-1]%(n+1)==0, ('counterexample',n+1)
    if (n+1)%1000==0: print('Exact original sums checked through',n+1,'elapsed',round(time.
    monotonic()-start,2),flush=True)
```

```
def prime(p): return p>=2 and all(p%d for d in range(2,isqrt(p)+1))
def valuation(n,p):
    v=0
    while n%p==0: v+=1; n//=p
    return v

double=[]; split=[]; double_count=split_count=0
for p in range(3,N+1,2):
    if not prime(p): continue
    for m in range(1,N//(p*p)+1):
        mod=p**(valuation(m,p)+3)
        double_count+=1
        if (S[p*p*m]-p*p*S[m])%mod: double.append((p,m,mod))
    if p%4==1:
        for m in range(1,N//p+1):
            mod=p**(valuation(m,p)+2)
            split_count+=1
            if (S[p*m]-p*S[m])%mod: split.append((p,m,mod))

result={'n_max':N,'original_divisibility_failures':[],
        'double_lift_count':double_count,'double_lift_failures':double,
        'split_lift_count':split_count,'split_lift_failures':split,
        'elapsed_seconds':round(time.monotonic()-start,3),
        'a_sha256':hashlib.sha256(('\n'.join(map(str,a))+'\n').encode()).hexdigest()}
print(json.dumps(result,indent=2),flush=True)
Path(__file__).with_name('exact_lift_test_results.json').write_text(json.dumps(result,indent=2)+
    '\n')
```

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